Yufan Du

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Education

Peking University *Sep. 2021 - Present*
Bachelor of Science in Applied Physics *and* Computer Science Dual Major *Beijing, China*

Relevant Scores

Overall GPA:	3.850/4.000 (3^{rd} out of 64 students)
TOEFL:	108 (R27 + L30 + S23 + W28)
GRE:	324 + 4.0

Technical Skills

Knowledgeable with: C/C++, Python
Familiar with: Python ML libraries (PyTorch, NumPy), Verilog, FPGA, Embedded systems, Linux systems, MATLAB, Tcl, LaTeX, SQL/Database, JavaScript/HTML
EDA tools: Familiar with Cadence Innovus, Cadence Virtuoso

Awards and Honors

ICCAD'24 Contest First Prize (1/45) (Team Leader)	ASU and NVIDIA	2024
Best Paper Award Nomination	ICCAD	2024
Industrial and Commercial Bank of China (ICBC) Rising Star Scholarship (1%)	ICBC	2024
Academic Innovation Award	Peking University	2024
Outstanding Student Leader Award (3%)	Peking University	2024
Huatai Securities Technology Scholarship	Huatai Securities	2023
Outstanding Student Leader Award (3%)	Peking University	2023
Huatai Securities Technology Scholarship	Huatai Securities	2022
Outstanding Student Award (10%)	Peking University	2022

Publications

- **Yufan Du**, Zizheng Guo, Yibo Lin, Runsheng Wang and Ru Huang, "Fusion of Global Placement and Gate Sizing with Differentiable Optimization," *2024 International Conference on Computer-Aided Design (ICCAD24)*. (**Best Paper Award Nomination**)
- Zizheng Guo, Zuodong Zhang, Wuxi Li, Tsung-Wei Huang, Xizhe Shi, **Yufan Du**, Yibo Lin, Runsheng Wang and Ru Huang, "HeteroExcept: A CPU-GPU Heterogeneous Algorithm to Accelerate Exception-aware Static Timing Analysis," *2024 International Conference on Computer-Aided Design (ICCAD24)*.
- **Yufan Du**†, Zizheng Guo†, Xun Jiang, Zhuomin Chai, Yuxiang Zhao, Yibo Lin, Runsheng Wang and Ru Huang, "PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning," *2024 Design Automation Conference (DAC24)*.

Research Experiences

Differentiable Cirtut Logic Optimization *Sep. 2024 - Present*

- Advisor: Professor Yibo Lin
- Investigate the application of differentiable optimization in logic optimization.

Neural Architecture Search (NAS) for Circuit Design *Jun. 2024 - Present*

- Advisor: Professor David Z. Pan and Yibo Lin
- Investigate the application of NAS in circuit design optimization (circuit transformation).
- *Ready for submission to DAC 2025.*

CUDA-Accelerated Gate Sizing *Jun. 2024 - Sep. 2024*

- Advisor: Professor David Z. Pan.
- Leverage CUDA programming to accelerate gradient descent method-based gate sizing.
- *First Prize in ICCAD 2024 Contest Problem C.*

Gate Sizing and Global Placement Co-optimization *Dec. 2023 - May. 2024*

- Advisor: Professor Yibo Lin.
- Unlock larger optimization space.
- Enable GPU-accelerated gradient descent with differentiable PPA objectives.
- Achieve 77.1% TNS and 43.5% WNS improvement with 1% lower power consumption.
- Speed up the process by up to 7× faster than traditional flow.
- *Accepted by ICCAD 2024 (Best Paper Award Nomination).*

Machine Learning for Chip Power Prediction *Jun. 2023. - Nov. 2023*

- Advisor: Professor Yibo Lin.
- Early-stage power prediction to guide design optimization.
- Model cell power and clock power separately + module-wise power calibration.
- Achieve 2% relative error on average for total power prediction.
- *Accepted by DAC 2024.*

Project Experiences

FPGA-Based Gesture-Controlled Snake Game Design *Mar. 2023 - Jul. 2023*

- Advisor: Professor Xiaohui Duan.
- Deploy *computer vision algorithms* (from Canny edge detection, circular hough transformation, to pattern recognization) and the classic Snake game on a resource-limited FPGA platform.
- Develop and implement a *real-time control mechanism* with a camera detecting gestures.
- Provide insights for *IoT edge device*.

AI-Assisted Schedule Manager *Jan. 2023 - Apr. 2023*

- Advisor: Professor Wei Guo.
- Develop a *cross-platform application* integrated with commercial AI API for users' agenda scheduling.
- Provide better user experience through conversational interfaces for management and reminders.